

EAM PRO

Enterprise Asset Management Platform

Streamline maintenance operations, maximize asset uptime, and drive operational excellence with intelligent automation.

40%

Reduced
Downtime

25%

Cost Savings

30%

Productivity Gain

Product Overview | 2026

Confidential | For Client Review Only

What We'll Cover Today

1 Platform Overview & Architecture

Technology stack, design philosophy, and deployment

2 Asset Management

Complete asset lifecycle from onboarding to retirement

3 Complaint & Job Order Workflow

End-to-end service management with technician assignment

4 Preventive Maintenance

Automated scheduling, mileage-based triggers, and inspection checklists.

5 Rental & Leasing Operations

Contract management, billing, equipment availability

6 Inventory & Spare Parts

Stock management, demand forecasting, supplier tracking

7 FRACAS & Quality Assurance

Failure analysis, root cause identification, corrective actions

8 Reporting & Analytics

MTBF/MTTR, dashboards, performance metrics

9 Financial Management

Invoicing, payments, cost tracking

10 Security, Compliance & Integrations

RBAC, audit trails, WhatsApp, email notifications

Built for Enterprise Scale & Reliability

1 Backend

- Flask (Python 3.12+)
- SQLAlchemy 2.x ORM
- Marshmallow serialization
- RESTful API (200+ endpoints)
- JWT authentication

2 Front end

- Bootstrap 5 responsive UI
- Chart.js visualizations
- Jinja2 templating
- Mobile-first design
- Font Awesome icons

3 Data base

- PostgreSQL
- 40+ data models
- Optimized queries
- Automated backups
- Data integrity constraints

4 Security

- Role-based access (RBAC)
- Encrypted passwords
- Complete audit trail
- Session management
- IP tracking

200+

API Endpoints

40+

Data Models

70+

UI Pages

12+

Core Modules

Complete Asset Lifecycle Management

1

Asset Registry

Serial number & specification tracking
Make, model & OEM management
QR/Barcode identification Photo &
document attachments

2

Operations

Mileage & meter reading tracking
Commission date management
Status lifecycle management
Running hours monitoring

3

Classification

Asset types & categories
Purchase conditions (New/Refurbished/Leased)
Ownership models (Self/Customer/Partner)
Multi-site location tracking

4

Data Management

Excel bulk import/export
Asset assignment tracking
Maintenance history linkage
Complete audit trail

Unified Stakeholder Database

1 Partner Types

Employees & Technicians
Customers & Clients
OEM Vendors
Service Vendors & Suppliers

2 Contact Management

Phone, email & address,
multi-site locations per
partner, hierarchical
manager structure,
department organization.

3 Site Management

Multiple locations per
customer, site-specific asset
tracking, geographic
organization, location-based
service.

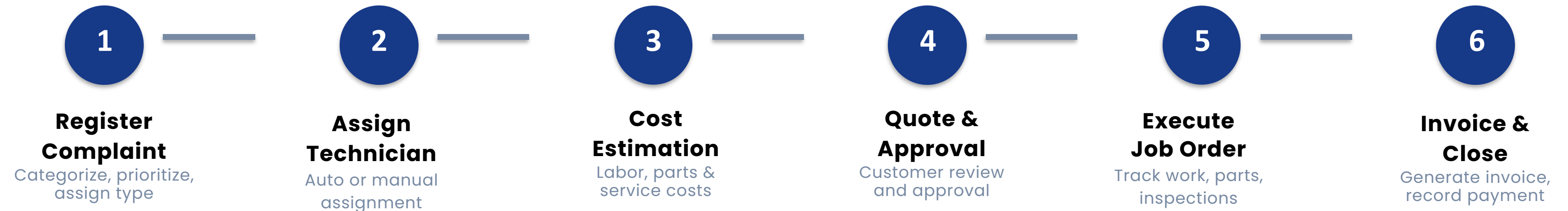
4 Integration

User account linking Job
order assignment Invoice
association Complete
activity audit trail

Unified Partner Model

A single, flexible model manages all stakeholders—employees, customers, vendors, and suppliers—reducing data duplication and enabling seamless cross-module integration across complaints, job orders, invoices, and rental contracts.

Complaint to Closure – End to End



1 Complaint Management

Types: Job Site, In-House, Warranty, Emergency
Priority: Low, Medium, High, Critical
Auto-assignment based on rules
10-step status workflow with full tracking
Notification system for deadlines & changes

2 Job Order Execution

Primary + supporting technician assignment
Spare parts tracking & costing
Inspection checkpoints (Pass/Fail/NA)
Pause/Resume with documented reasons
Time tracking & photo evidence

Proactive Maintenance, Minimized Downtime

Master Maintenance Plans

Reusable templates with predefined tasks, checkpoints, and schedules

Flexible Scheduling

Daily, weekly, monthly, quarterly, yearly frequency options

Mileage-Based Triggers

Automatic work orders generated based on meter readings

Inspection Checklists

Configurable checkpoints with expected values and pass/fail criteria

Auto-Complaint Creation

System automatically creates complaints when maintenance is due

Calendar View

Visual scheduling with due date tracking and overdue indicators

Workflow: Create Plan › Define Checkpoints › Assign to Asset › Auto-Schedule › Generate Complaint › Execute & Verify

Complete Rental Lifecycle Management



1

Contract Management

Multiple equipment per contract, flexible pricing (daily/weekly/monthly/yearly), payment terms configuration, agreement document generation, equipment condition tracking.

2

Equipment Availability

Buckets: in yard, on rent, in shop; real-time status dashboard; reserved & sold tracking; utilization metrics & revenue.

3

Rental Invoicing

Auto-invoicing by billing frequency, multi-line invoice support, professional PDF generation, payment tracking & aging, overdue alerts & reminders.

4

Notifications

Contract ending alerts
Pending delivery reminders
Unpaid invoice notifications
Real-time bell notification
Meter reading overage alerts

Smart Spare Parts & Materials Tracking

1 Material Master

Complete parts catalog
Category & type classification
Supplier linkage
Specifications & pricing
Minimum stock levels

2 Inventory Movements

Stock in/out tracking
Movement history log
Job order consumption
Transfer between locations
Real-time stock levels

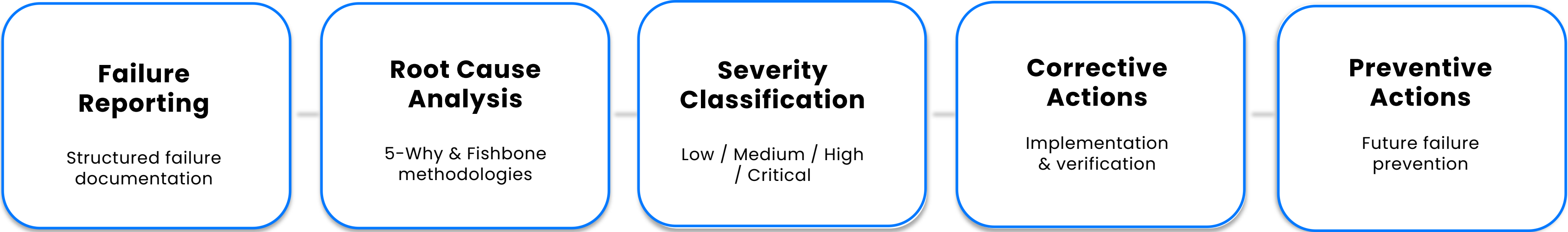
3 Demand Forecasting

Predictive spare demand analysis
Historical consumption patterns
Seasonal trend identification
Reorder point optimization
Cost projection & budgeting

Spare Demand Forecasting

AI-powered predictive analytics analyzes historical consumption patterns to forecast spare part demand, optimize reorder points, and reduce stockout risk while minimizing carrying costs.

Root Cause Analysis & Continuous Improvement



1

Inspection Checkpoints

Configurable checklist items

Pass / Fail / N/A results

Expected value comparison

Manager review workflow

Photo evidence attachment

2

Continuous Improvement

Failure pattern identification

Trend analysis across assets

Corrective action verification

Prevention strategy tracking

Complete investigation audit trail

Data-Driven Decisions at Every Level

Reliability Analysis

MTBF & MTTR

Mean Time Between Failures and Mean Time To Repair tracking for reliability metrics

Asset Reports

Performance Tracking

Health scores, performance trends, and complete asset history analysis

Technician KPIs

Workforce Analytics

Productivity metrics, job completion rates, and workload distribution

Failure Analysis

FRACAS Reports

Failure patterns, root cause trends, and corrective action effectiveness

Maintenance Reports

Schedule Compliance

PM adherence, cost analysis, work order metrics, and backlog tracking

Financial Reports

Cost & Revenue

Maintenance spending, rental revenue, ROI analysis, and budget tracking

Operational Intelligence at a Glance

Total Assets

Track All
Equipment

Active Jobs

Real-Time Work
Orders

Technicians

On-Duty Staff

Monthly Cost

Maintenance Spend

**Equipment on
Rent**

Active Rentals

Overdue Alerts

Attention Required

1 Chart Visualizations

Asset performance trends
(week/month/year)
Maintenance cost breakdown
Work order status distribution
Equipment utilization pie charts
Technician workload bar graphs

2 Interactive Features

Date range filtering
Module-specific drill-downs
Real-time data refresh
CSV/Excel export capability
Print-ready report views

3 Alerts & Notifications

Overdue maintenance warnings
Contract expiration alerts
Low stock notifications
Unpaid invoice reminders
Bell notification center

Invoicing, Payments & Cost Control

1 Cost Estimation

- Labor cost calculation
- Spare parts costing
- Service cost estimation
- Multi-line item quotes
- Customer approval workflow

2 Job Order Invoicing

- Auto-populate from joborders
- Line item management
- Configurable tax rates
- Discount support
- Professional PDF generation

3 Rental Invoicing

- Auto-generate by billing cycle
- Daily/weekly/monthly/yearly
- Meter overage charges
- Damage assessment billing
- Multi-equipment invoices

4 Payment Tracking

- Draft > Sent > Paid workflow
- Partial payment recording
- Balance tracking per invoice
- Overdue payment alerts
- Complete payment history

Financial Workflow

EstimateCosts > Generate Quote > Customer Approval > Execute Work > Create Invoice > Send to Customer > Record Payment > Close

Enterprise-Grade Security & Connectivity

1 Access Control

Role-Based Access (RBAC)
Granular module
permissions Data scoping
(all vs. allotted) Multi-role
user support Secure session
management

2 Audit & Compliance

100% activity traceability
User attribution on all
actions IP address tracking
Timestamped audit logs
Regulatory compliance
reports

3 Integrations

WhatsApp Business API
SMTP email notifications
Excel/CSV import & export
ZPL label printer support
Barcode/QR code
generation

4 Data Security

Encrypted password storage
JWT token authentication
PostgreSQL database security
Automated database
backups Secure file storage

Highly Configurable

Custom reference prefixes | Multi-currency support | Timezone configuration | Date format localization | Company branding | Lookup value management

What Sets EAM Pro Apart

Pause/Resume Jobs

Track work interruptions with documented reasons for accurate time & productivity analysis

Mileage-Based PM

Automatic maintenance triggers based on meter readings, not just calendar schedules

Equipment Availability Dashboard

Real-time visibility into every asset. In use, on rent, in shop, Reserved, or Sold

Demand Forecasting

Predictive spare parts analytics to optimize inventory levels and reduce stockouts

Multi - Technician Assignment

Assign primary and supporting technicians with independent time tracking

Integrated Rental + Maintenance

Unified platform handling both rental operations and maintenance in one system

WhatsApp + Email Notifications

Reach technicians and customers through WhatsApp Business API and configurable SMTP

Complete Financial Workflow

From cost estimation through quote approval to invoicing and payment tracking



EAM PRO

Thank You

Ready to transform your
maintenance operations?

NEXT STEPS

Detailed module walkthrough & live demo
Custom configuration for your workflow
Pilot deployment & user training
Go-live support & ongoing maintenance

12+
Core Modules

200+
API Endpoints

40+
Data Models

70+
UI Pages